

Open X-Embodiment · arXiv 2310.08864 · Nov 2 · Data

Archaeologist

11+3

RoboNet/Bridge/RT-1 → OXE+RT-X
→ Octo/OpenVLA soups → π_0 /GR00T → BEHAVIOR

Paper club · Data cluster · due 9:30 AM PT

Spine: pool $\neq$ train mix · capacity $\times$ mix · steal mix science, don't drink the river

Open X-Embodiment Collaboration · 21 institutions · robotics-transformer-x.github.io

Archaeologist talk arc · 11+3



Spine: pool $\neq$ train mix · capacity $\times$ mix · steal mix science, don't drink the river

Genealogy — OXE → soups → BEHAVIOR



What OXE replaced

Single-lab teleop as default pretrain
“One robot, one policy” as only recipe

What OXE did NOT replace

Post-train gold demos · wrist/proprio
Continuous chunks · challenge sim alignment

Children drink the water: Octo compares to RT-1-X · OpenVLA beats RT-2-X on open suite

π_0 open share $\approx$ 9.1% OXE Magic Soup + Bridge + DROID · BEHAVIOR needs OXE-scale thinking, not raw dump

CLEAR: Pool ≠ RT-X train mix

Marketing blur is the #1 citation failure mode

OXE POOL

Datasets	60
Embodiments	22
Institutions / labs	21 / 34
Trajectories	1M+
Skills / tasks	527 / 160,266
Scenes	311
Schema	RLDS TFRecords

RT-X TRAIN MIX

Manipulators	9 (≠ full 22)
Sources (ex.)	RT-1, QT-Opt, Bridge, Play, Jaco, Cable, ...
RT-1-X	robotics mix only
RT-2-X	~1:1 web : robotics
Eval	3600 trials / 6 robots
Downstream	Octo ~800k · OpenVLA ~970k

Moral: every claim needs a frozen mix card — pool size ≠ train size ≠ Octo/OpenVLA subset

Cite “22 robots / 1M+” for the commons; cite “9 manipulators” for RT-X experiment results

Schema / consolidation used for RT-X

Observation

Image history + language; one canonical camera per dataset; unaligned poses

Action

7-DoF EE (x,y,z,r,p,y,grip) — abs/rel/vel as in source; NOT frame-aligned

Discretization

256 bins/dim · 8 outputs = 7 EE + episode terminate

Normalization

Per-dataset before binning; de-norm is embodiment-specific

Gap module

NONE — shared policy eats the soup; transfer is learned

RT-1-X

35M FiLM-EfficientNet + Transformer · robotics mix only · 3-10 Hz local

RT-2-X

PaLI-X VLA · actions as text tokens · ~1:1 web:robot · 5B/55B

Open X-Embodiment: Robotic Learning Datasets and RT-X Models

Open X-Embodiment Collaboration⁰

robotics-transformer-x.github.io

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Fig. 2 — Embodiments, scenes, trajectories, skill long-tail

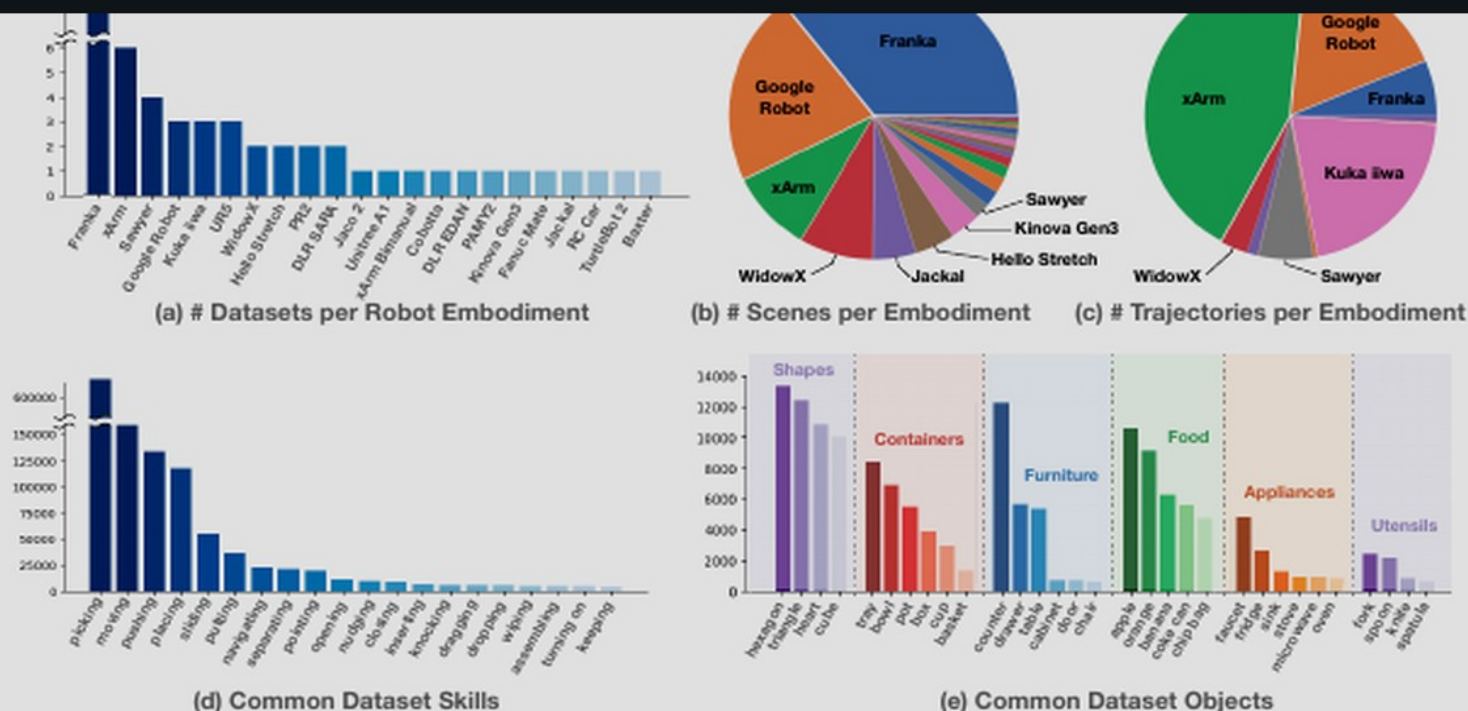


Fig. 2: The Open X-Embodiment Dataset. (a): the dataset consists of 60 individual datasets across 22 embodiments. (b): the Franka robot has the largest diversity in visually distinct scenes due to the large number of Franka datasets, (c): xArm and Google Robot contribute the most number of trajectories due to a few large datasets, (d, e): the dataset contains a great diversity of skills and common objects.

prior works, we directly train a policy on X-embodiment data, without any mechanisms to reduce the embodiment gap, and observe positive transfer by leveraging that data.

Large-scale robot learning datasets. The robot learning community has created open-source robot learning datasets, spanning grasping [60–71], pushing interactions [23, 72–74], sets of objects and models [75–85], and teleoperated demonstrations [8, 86–95]. With the exception of RoboNet [23], these datasets contain data of robots of the same type, whereas we focus on data spanning multiple embodiments.

- **Open X-Embodiment Dataset:** robot learning dataset with *1M+* robot trajectories from 22 robot embodiments.
- **Pre-Trained Checkpoints:** a selection of RT-X model checkpoints ready for inference and finetuning.

We intend for these resources to form a foundation for X-embodiment research in robot learning, but they are just the start. Open X-Embodiment is a community-driven effort, currently involving 21 institutions from around the world, and we hope to further broaden participation and grow the

Fig. 3 — RT-1-X / RT-2-X architecture + discrete EE

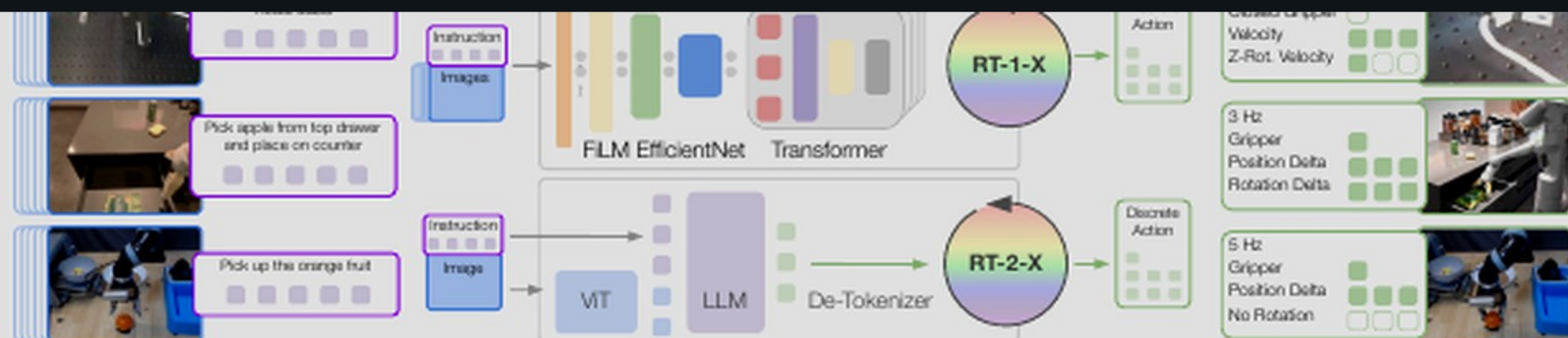


Fig. 3: RT-1-X and RT-2-X both take images and a text instruction as input and output discretized end-effector actions. RT-1-X is an architecture designed for robotics, with a FiLM [116] conditioned EfficientNet [117] and a Transformer [118]. RT-2-X builds on a VLM backbone by representing actions as another language, and training action text tokens together with vision-language data.

metadata) per embodiment (Fig. 2(b)), where Franka dominates. Fig. 2(c) shows the breakdown of trajectories per embodiment. To further analyze the diversity, we use the language annotations present in our data. We use the PaLM language model [3] to extract objects and behaviors from the instructions. Fig. 2(d,e) show the diversity of skills and objects. While most skills belong to the pick-place family, the long tail of the dataset contains skills like “wiping” or “assembling”. Additionally, the data covers a range of household objects, from appliances to food items and utensils.

IV. RT-X DESIGN

To evaluate how much X-embodiment training can improve the performance of learned policies on individual robots, we require models that have sufficient capacity to productively make use of such large and heterogeneous

B. Policy architectures

We consider two model architectures in our experiments: (1) RT-1 [8], an efficient Transformer-based architecture designed for robotic control, and (2) RT-2 [9] a large vision-language model co-fine-tuned to output robot actions as natural language tokens. Both models take in a visual input and natural language instruction describing the task, and output a tokenized action. For each model, the action is tokenized into 256 bins uniformly distributed along each of eight dimensions; one dimension for terminating the episode and seven dimensions for end-effector movement. Although both architectures are described in detail in their original papers [8, 9], we provide a short summary of each below:

RT-1 [8] is a 35M parameter network built on a Transformer architecture [118] and designed for robotic control,

Headline numbers (tattoo)

Fig. 4

~+50%

relative mean SR lift
small-data domains
RT-1-X vs Original / RT-1

Table I

73% vs 92%

RT-1-X underfits
Google Robot 6 skills
capacity × mix

Table II

27.3 → 75.8

emergent skills ~3×
Bridge→Google Robot
X-mix transfer

Ablation

42.8%

no-Bridge mix
emergent collapses
Bridge is load-bearing

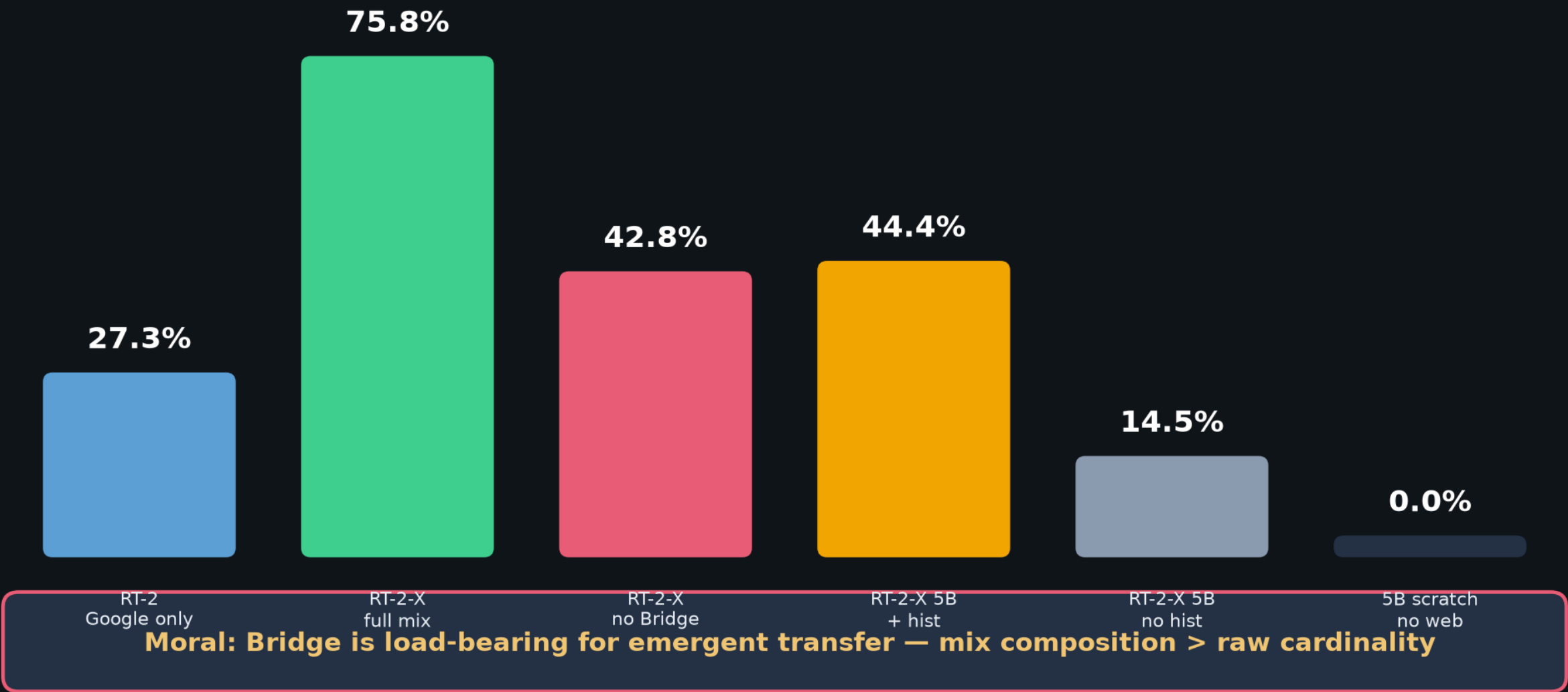
Table I — large native domains (exact)

Setting	Original	RT-1	RT-1-X	RT-2-X 55B
Bridge IRIS WidowX	13%	40%	27%	50%
Bridge RAIL WidowX	13%	30%	27%	30%
Google 6 skills	—	92%	73%	91%

RT-1-X hurts large in-domain SR — positive transfer is conditional on capacity

Table II — Bridge ablation (emergent skills)

Bridge→Google Robot skills absent from Google-only data



Anti-patterns → fixes for BEHAVIOR

OXE-era raw practice → later soup science → your action

OXE-era fail	Fix to copy	BEHAVIOR action
Uniform everything	Curated weight table	Explicit mix card
Loyalty to bad shards	Kill switch → weight 0	Drop when loss stalls
Wrist poverty	Hard gate contact tasks	No wrist → down-weight
Lang poverty	Dual-condition + subgoals	Lang $\geq 90\%$ post-train
7×256 bins only	Chunked continuous / FAST	Not naïve discrete EE
Huge shards drown gold	Sublinear $n^{0.43}$	Protect ~20k gold demos
Pick-place monopoly	Skill-family caps	Cap family $\leq \sim 20\%$

Red lines + municipal water metaphor

OXE = MUNICIPAL WATER

Shared multi-lab supply
Enabled RT-X / Octo / OpenVLA / π_0
Agglomeration + RLDS schema
Positive transfer existence proof

BEHAVIOR'26 = BOTTLED BLEND

Specialty challenge mix
Stage-rich · wrist-rich · schema-clean
~20k gold demos as post-train depth
Steal science — don't drink the river

DO NOT

- × Dump raw OXE into BEHAVIOR FT at uniform weight
- × Assume discrete 7×256 bins are enough for 2026 long-horizon household
- × Claim “OXE alone = BEHAVIOR'26 data plan”
- × Cite unpublished OXE hours
- × Skip mix-card discipline / transfer ablations

OXE replaced single-lab pretrain assumption
Did NOT replace post-train demos / wrist / chunks / sim align

RoboNet→Bridge/RT-1 → OXE+RT-X → Octo/OpenVLA → π_0 /GR00T → BEHAVIOR'25-'26

Q&A · keep the spine clear

Open X-Embodiment

Robotic Learning Datasets and RT-X Models

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OXE = municipal water · BEHAVIOR'26 = bottled specialty blend

Steal mix science — don't drink the river

Pool: 60 ds / 22 emb / 1M+ traj Train mix: 9 manipulators Eval: 3600 trials / 6 robots

Headlines: Fig.4 ~+50% · Table I RT-1-X 73% vs RT-1 92% · emergent 27.3%→75.8% · no-Bridge 42.8%

Seat 1 Visionary = Kanta Saito · Quiet build